

Numerical research on the flow and heat transfer characteristics in the immersion jet cooling for servers[☆]

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ABSTRACT

Immersion cooling and jet impingement liquid cooling, as two of the significant emerging direct contact liquid cooling methods, have promising prospects for development and application. The two cooling methods are combined and numerical simulation is applied to study the heat transfer performance of the immersion jet-cooled server. The orthogonal method is employed to determine the optimal structural parameters of the immersion jet, and the flow and heat transfer characteristics under different inlet and outlet layouts are investigated. The study shows that the optimal heat dissipation effect and moderate pressure are achieved when the ratio of jet distance (L) to jet pipe diameter (D) is 1.2, and the ratio of jet hole spacing to jet orifice diameter is 11.1. When the inlet and outlet of the cabinet are arranged in a Z-shaped layout, the pressure on the server surface is reduced by 17.1 %–31.6 % compared to a U-shaped layout. When the inlet and outlet of the cabinet are arranged in a Z-shaped manner with the inlet at the bottom and the outlet at the top, the highest heat transfer coefficient is over $24000 \text{ W}/(\text{m}^2 \cdot \text{K})$, which is approximately 1.5 times higher than that of other layouts. Through dimensional analysis, a quadratic function was selected to fit the relationship between heat transfer coefficient h , server pressure P and jet ratio (D/L), opening area A ; In addition, the criterion relationship of $Nu = f(Re, Eu)$ is fitted according to the simulation data, which provides a theoretical basis for energy saving and consumption reduction in data centers.

1. Introduction

As of the year 2023, the nationwide data center rack scale has reached 7.6 million units. The Huajing Industry Research Institute predicts that by the year 2025, the national data center power consumption will reach a new high of 395.2 billion kW-h, with the energy consumption for electronic equipment cooling accounting for 38 % of the total energy consumption [1]. According to data released by Colocation America, the average power per rack reached 16.5 kW in 2020, a 175 % increase compared to 2008, and it is expected to reach 25 kW per rack by the year 2025.

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As the power density of individual racks continues to increase, the heat flux density of data centers also increases. Data centers with a heat flux density ranging from 100 to 1000 W/cm² are typically referred to as high heat flux density data centers [2]. Furthermore, the stability of electronic components is greatly affected by temperature, with a 10 °C increase in temperature leading to a 5 % decrease in component reliability when the temperature reaches 70–80 °C [3]. Conventional air-cooling heat exchange systems are increasingly unable to meet the cooling requirements of high heat density servers, making the pursuit of more efficient cooling systems the current focus of research.

Liquid cooling has garnered widespread attention due to its strong heat transfer capability, good thermal stability, high adaptability, and low energy consumption [4–7]. With the increasing integration of servers in data centers, liquid cooling technology is gradually becoming the solution for large-scale data centers. Immersion cooling and jet impingement cooling, like emerging direct contact liquid cooling methods that achieve efficient heat transfer, have attracted the attention of many scholars both domestically and internationally. Jet impingement cooling shares the same heat transfer mechanism as immersion cooling and represents an upgraded version of the latter [8].

With the increasing scale and integration of data centers, particularly since the 21st century and especially since 2010, there has been a surge of research interest in immersion liquid cooling technology in the electronic device industry, including data centers. Wagner [9] compared the cooling performance of three different liquid cooling technologies. The experimental results indicated that the single-phase mineral oil immersion cooling had the lowest power consumption and the lowest thermal resistance. Yong [10] conducted an energy and performance comparison of an advanced air-liquid hybrid cooling and an immersion direct liquid cooling system. The results showed that the immersed liquid cooling solution achieved a PUE of 1.14 at full load, while the air-liquid hybrid system had a PUE of 1.48. Matsuoka [11] studied the cooling effects of silicone oil, soybean oil, and electronic fluorocarbon liquid as coolants within the server cabinet using computational fluid dynamics. The simulation results indicated a close correlation between cooling effectiveness and the viscosity of the refrigerant. Gandhi [12] investigated the impact of different coolant temperatures and flow rates on the cooling performance of liquid-cooled servers. The results indicated that at a coolant temperature of 45 °C and a flow rate of 3.8 L/min, the CPU temperature could be maintained at approximately 70 °C. Cheng [13] developed a single-phase immersion server model and conducted simulations for different coolant flow velocities. The results showed that higher flow velocities resulted in greater heat dissipation and a more uniform temperature distribution.

Ma [14,15] conducted experiments using circular nozzles to produce jets with diameters ranging from 500 µm to 100 µm and slot jets with nozzle widths ranging from 90 µm to 540 µm. They tested the heat transfer characteristics of various coolants, including air, nitrogen, CO₂, water, etc. Hammad and Wang [16,17] employed PIV measurement technique to investigate the flow field structure of vertical impinging jets under different impingement heights and Reynolds numbers. They found that the time-averaged velocity is determined by the impingement height rather than the Reynolds number magnitude. Michna [18] studied the relationship between pressure drop and heat transfer coefficient in single-phase impinging jet. Their experimental setup could achieve a heat dissipation requirement of 400 W/cm² under a temperature difference of 50 °C. Ai [19] examined the influence of nozzle velocity on heat transfer rate under different heat fluxes and flow rates. The experimental results indicated that the moving nozzle performs better than the fixed nozzle in reducing the maximum temperature difference and average liquid film thickness on the heated surface, exhibiting a more stable heat transfer rate.

Despite the strong convective heat transfer capability of a single-nozzle jet, the significant gradient in the convective heat transfer coefficient can still lead to an excessively large surface temperature difference on the cooled surface, which is not beneficial for the surface stability. Therefore, in engineering practice, the application of array jet cooling [20–22] is more prevalent. Consequently, numerous scholars have initiated investigations into the heat transfer characteristics of array jet cooling. Bandhauer [23] proposed an array jet cooling device. This device could cool a surface with dimensions of 2 cm × 2 cm and a heat power of 165 W. Under a nozzle Reynolds number of only 305, the convective heat transfer coefficient reached 13100 W/(m²·K). The research findings of Devahdhanush [24] demonstrated that, when the jet velocities are equal, the critical heat flux density of array jet cooling surpassed that of single-hole jet cooling. Ekkad [25] compared the heat transfer effects of staggered and inline arrangements, revealing that the heat transfer performance of the staggered arrangement surpassed that of the inline arrangement. In experiments, Fabbri [26] observed a considerable pressure drop associated with array jet cooling, particularly pronounced at small spacings and aperture sizes.

Both immersion cooling and jet impingement cooling are emerging and highly efficient cooling technologies. Although immersion liquid cooling and jet impingement cooling technologies have achieved success in their respective fields, there is currently limited research on the combined liquid cooling approach. Existing studies lack a comprehensive consideration of the cooling effect of the combined action of these two cooling options. Additionally, researches on jet cooling had mostly concentrated on gas cooling, with less attention paid to the flow and heat transfer studies of liquid jet cooling.

The novelty of this paper is to improve the heat dissipation efficiency and stability of the system by integrating immersion cooling technology with jet cooling technology. In this paper, FLUENT software is used for numerical simulation. Orthogonal arrays are employed to conduct grouped simulations on structural parameters. Taking into consideration both heat transfer effectiveness and pressure drop, orthogonal analysis is conducted to seek the optimal structural parameters for immersion jet cooling. Furthermore, by altering the arrangement of the immersion liquid inlet and outlet, methods using structural changes are investigated to enhance server cooling effectiveness and temperature uniformity, with the aim of providing theoretical and practical guidance for the further development of server cooling technology.

2. Numerical simulation

2.1. Geometric modelling

The two-dimensional single-phase immersion jet liquid cooling cabinet model, as illustrated in Fig. 1, primarily consists of three components: server, jet pipe, and the immersion region.

The immersion region comprises an inlet pipe and an outlet pipe, with server vertically inserted into the cooling liquid. Considering the complexity of electronic component models within the server, the entire model underwent simplification. The server has dimensions of 150.8 mm in height and 85 mm in width, with jet holes evenly distributed along the jet pipe. The distance from the jet pipe to the server is denoted by L , which is 8 mm, 10 mm and 12 mm, the diameter of the jet pipe is denoted by D , which is 8 mm, 10 mm and 12 mm, and the diameter of the jet hole is denoted by d , which is 1.8 mm, 2.5 mm and 3 mm.

2.2. Numerical model

To simplify the calculations, the following assumptions are carried out about the model:

- (1) The flow is characterized by two-dimensional, steady, and incompressible conditions.
- (2) The flow regime is characterized by steady turbulence.
- (3) The impact of thermal radiation is neglected.
- (4) The dissipation heat caused by the work done by viscous forces in the fluid is disregarded.

Based on the above assumptions and guided by the principles of mass conservation, momentum conservation, and energy conservation, the governing equations for the fluid flow and heat transfer within the cabinet include continuity equation, momentum equation, and energy equation.

2.2.1. Governing equations

- (1) Continuity equation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0 \quad (1)$$

- (2) Conservation of momentum equation:

$$\frac{\partial(\rho u)}{\partial \tau} + \frac{\partial(\rho u^2)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \rho g_x \quad (2)$$

$$\frac{\partial(\rho v)}{\partial \tau} + \frac{\partial(\rho uv)}{\partial x} + \frac{\partial(\rho v^2)}{\partial y} = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + \rho g_y \quad (3)$$

- (3) Conservation of energy equation:

$$\frac{\partial t}{\partial \tau} + u \frac{\partial t}{\partial x} + v \frac{\partial t}{\partial y} = \frac{\lambda}{\rho c_p} \left(\frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} \right) \quad (4)$$

In the equation:

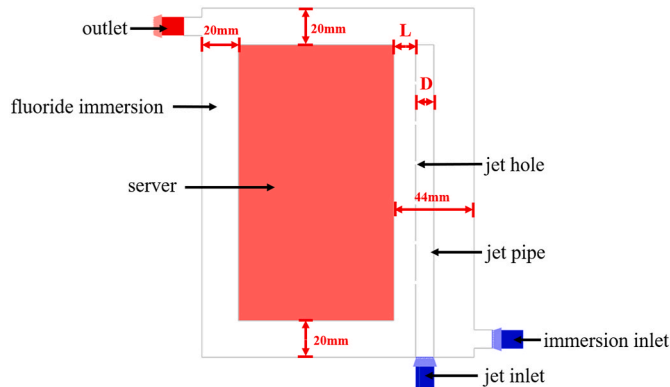


Fig. 1. Two-dimensional single-phase immersion jet liquid cooling cabinet model.

x, y -coordinates of a point in space;
 P, ρ, t -pressure, density, and temperature at that point;
 u, v -two directional velocities at the spatial point;
 μ, c_p -dynamic viscosity and specific heat capacity at that point.

2.2.2. Boundary conditions

The boundary conditions for this model are listed in Table 1 based on current operational data.

2.2.3. Solution methods

The velocity of the jet is generally high, indicating turbulent flow, thus necessitating the adoption of a turbulence model [27,28]. In this study, the standard *k-epsilon* model was selected due to its stability, cost-effectiveness, and high computational accuracy. Transient calculations are conducted to simulate the flow and heat transfer processes of the coolant immersed within the jet for 180 s. The pressure is corrected using the SIMPLE algorithm. The PRESTO discrete format was chosen for the pressure equation, and the second-order windward discrete format was used for the momentum, energy, turbulent kinetic energy and energy dissipation rate equations.

The gravity option is turned on, the direction of gravity is the negative half-axis of the Y-axis, and the standard wall function is selected for near-wall surfaces. Select the pressure solver, with the relaxation factor at the default setting. The convergence criteria for the velocity components along the X and Y axes, as well as for the continuity equation, are set at 10^{-4} , while the convergence criterion for the energy equation is set at 10^{-6} . The standard initialization method is used, with the option to compute from all zones and to patch the temperature of the server.

2.3. Grid division and independence verification

A well-constructed computational grid is paramount for successful and accurate solutions [29]. Insufficient grid resolution may lead to inaccuracies in the computational results, while excessive grid refinement can result in wastage of computational resources. Therefore, the selection of an appropriate grid resolution is of the utmost importance [30]. In this study, the grid generation process was accomplished through Meshing in Workbench. The mesh was dominated by an unstructured quadrilateral dominant mesh with sizing of the inlet and outlet as well as the injection holes, and the near-wall surfaces of the server. To improve the computational accuracy, local mesh encryption was performed in the region of the model where heat exchange is frequent. The overall cell size of the grid is 0.5 mm and the grid size of the encrypted area is 0.1 mm.

A grid independence test was conducted to assess the influence grid quantity to the simulation results. As depicted in Fig. 2, six sets of grid configurations were chosen with grid counts of 84,384, 119,094, 183432, 237,641 and 321,692. Upon reaching a grid quantity of 183,432, further increments in grid density resulted in negligible changes in the monitored server temperature. This observation effectively mitigated the impact of grid quantity on the simulation outcomes, and the final grid model is presented in Fig. 3.

2.4. Validation of model

To verify the validity of the numerical simulation method, a numerical model was established using the numerical simulation method proposed in Section 2.2, based on the same geometry model and adiabatic boundary conditions as those described in reference [9]. Literature [9] employed a mineral oil single-phase immersion cooling method to cool the cooper block, with a dimension of 25.4 mm×25.4 mm×5 mm. The bath box has a dimension of 50 mm×50 mm×100 mm. The cooper block was cooled using 40 °C mineral oil during the experiment. The mineral oil used as the coolant has the properties in Table 2.

Based on the dimensions of the experimental setup in literature [9], the physical model for validation is shown in Fig. 4.

The model was validated for a total of three calculations with copper block heat source power of 50, 100 and 150 Watts respectively. Fig. 5 compares the numerical simulation results to the experimental results, with a maximum error of 8.8 %. Therefore, it can be concluded that the numerical calculation method presented is effective.

3. Submerged jet cooling orthogonal test

In multifactor simulation schemes, the orthogonal experimental method is a scientific approach [31]. The orthogonal experimental

Table 1
Summary of boundary condition settings.

Boundary name	Boundary condition
Jet pipe inlet	Velocity inlet, 2 m/s
Cabinet inlet	mass-flow outlet, 2 kg/s
Cabinet outlet	pressure outlet, inhibition of reflux
Cabinet walls and Jet tube walls	Stationary wall, fluid-solid coupling
Coolant physical properties	Fluorinated liquid at 25 °C coefficient of dynamic viscosity, 0.0019 Pa s density, 1830 kg/m ³ specific heat capacity, 1100 J/(kg·K) thermal conductivity, 0.066 W/(m·K)
Server	energy source term, 1254.8W
Flow channel material	aluminium

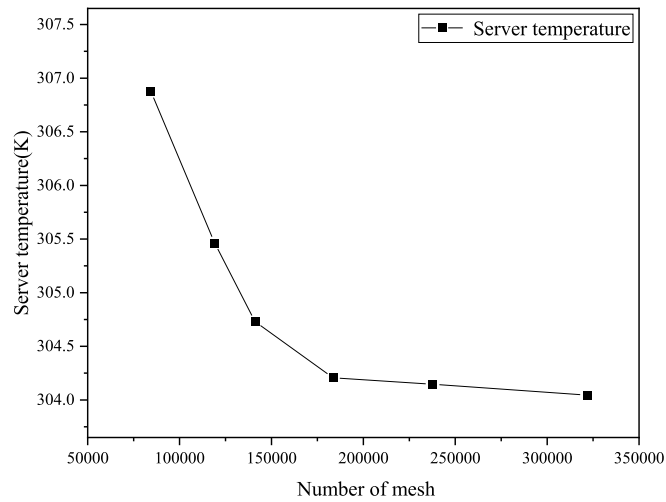


Fig. 2. Grid independence Verification.

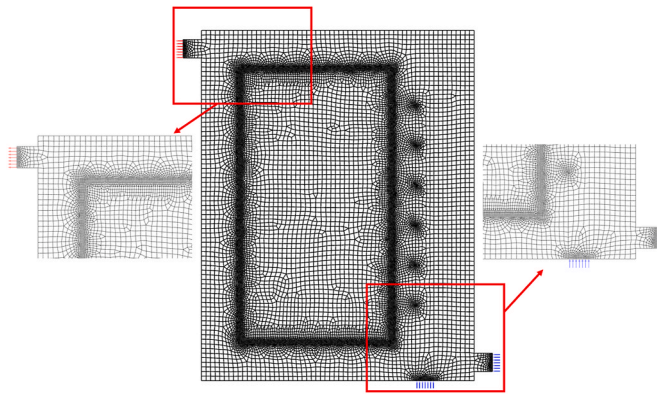


Fig. 3. Grid model diagram.

Table 2

Mineral oil properties.

Physical parameter	Value
thermal conductivity	0.13 W/(m · K)
viscosity	0.0379 N/m ³
specific heat	1927 J/(kg · K)
density	850 kg/m ³

method utilizes a systematic arrangement of orthogonal arrays to organize experimental groups. Orthogonal arrays possess characteristics of balanced dispersion and comparability, enabling the selection of optimal combination schemes based on a small number of simulated working conditions.

In the structural parameter design of submerged jet cooling devices, the values of parameters such as jet distance L , jet pipe diameter D , number of holes S , and jet orifice diameter d are specified within a given range. The optimal values can be obtained by orthogonal analysis within the parameter intervals.

In the immersed jet cooling device, the performance evaluation indicators of the device include the server's maximum temperature ($T_{\max}$ in °C), the inlet-outlet pressure difference (ΔP in Pa), the heat transfer coefficient (h in (W/(m²·K)), and the maximum pressure at characteristic points ($P_{\max}$ in Pa). The jet distance, nozzle diameter, number of nozzles, and nozzle diameter are the four indicators needed to optimize. The four indicators of jet distance L , jet pipe diameter D , number of jet holes S and diameter of jet orifice holes d are denoted by A, B, C and D, respectively. It is assumed that the above four factors do not affect or influence each other. Each factor is set at three levels, and an L_9 (3^4) orthogonal array is employed for experimental design. The values of the four factors at the three levels (For example, Factor A has a parameter level 1 of 8 mm, level 2 of 10 mm, and level 3 of 12 mm. Factors B, C and D are as well.) are

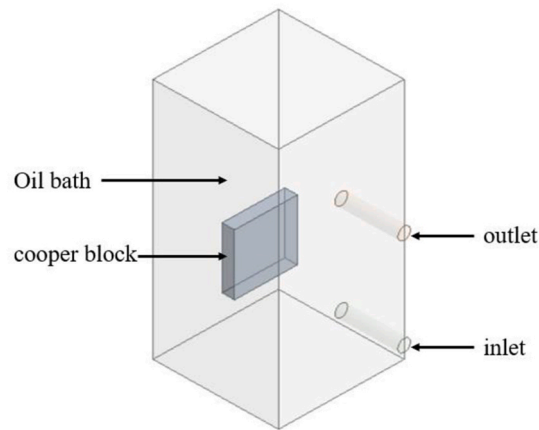


Fig. 4. Physical model.

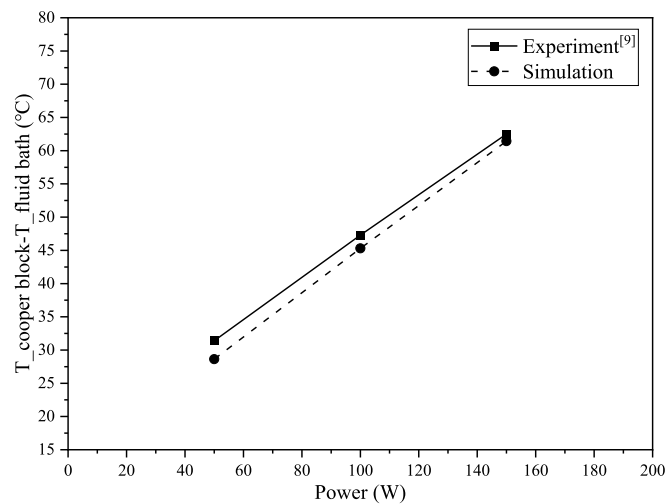


Fig. 5. Validation of the numerical calculation model.

presented in Table 3.

The three levels of each of the four factors are arranged into nine simulation groups as depicted in Table 4.

The values of the evaluation indexes in the orthogonal test are also needed to identify the characteristic points in the model that best reflect the values of the evaluation indexes. As shown in Fig. 6 below, the coolant flow rate is fast on the server wall near the jet pipe, and the pressure is correspondingly higher. Therefore, theoretically, the server wall near the jet pipe has the highest pressure of the whole server, so the point shown in Fig. 6 below is chosen as the characteristic point.

The results of heat transfer simulation for the submerged jet configurations are in Fig. 7 (where L8D8S6d1.8 indicates that the jet distance L is 8 mm, the jet pipe diameter D is 8 mm, the number of jet holes S is 6, and the diameter of jet orifice d is 1.8, and the rest of the numbering is thus). As can be seen in Fig. 7, temperature of the servers at a jet pipe diameter of 12 mm are lower than the temperature of the servers at a jet pipe diameter of 8 mm and 10 mm. However, based solely on the temperature distribution of the server, it cannot be concluded that a larger jet pipe diameter is the primary factor affecting the heat transfer efficiency of the

Table 3
Factor table for orthogonal tests.

Factor Parameter No.	Factor A	Factor B	Factor C	Factor D
	Jet distance	Jet pipe diameter D (mm)	Number of holes S	Jet orifice diameter d (mm)
	L (mm)			
1	8	8	6	1.8
2	10	10	7	2.5
3	12	12	8	3

Table 4
Simulation group.

Group No.	Factor A	Factor B	Factor C	Factor D
	Jet distance	Jet pipe diameter D (mm)	Number of holes S	Jet orifice diameter d (mm)
	L (mm)			
1	8	8	6	1.8
2	8	10	8	2.5
3	8	12	7	3
4	10	8	8	3
5	10	10	7	1.8
6	10	12	6	2.5
7	12	8	7	2.5
8	12	10	6	3
9	12	12	8	1.8

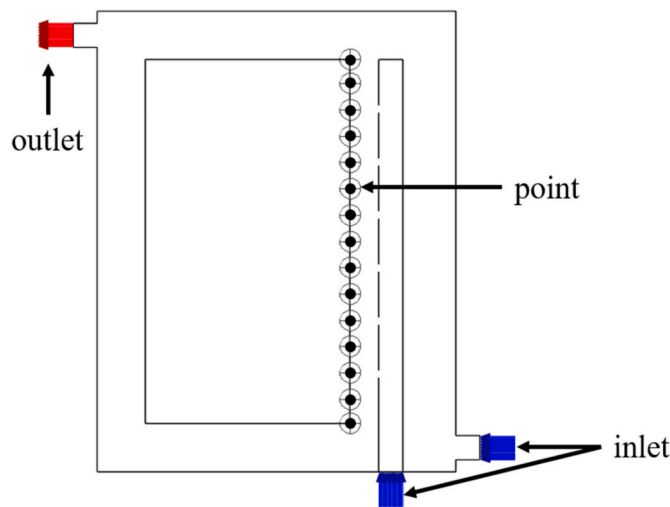


Fig. 6. Schematic of the location of the server feature points.

submerged jet system. The following analysis will proceed with orthogonal analysis to further investigate.

To precisely investigate the factors influencing the efficiency of submerged jet cooling, relevant pressure values were extracted from the designated characteristic points in Fig. 6, and server temperature values were obtained. The heat transfer coefficients were then calculated, and the resulting data are shown in Table 5 below.

The range analysis method consists of two steps: calculation and assessment. The calculation part includes the computation of K_{jm} , k_{jm} , and R_j , and the judgement part mainly includes the factors of primary and secondary judgement, superior level judgement and optimal combination judgement.

Among these, K_{jm} represents the sum of the experimental indicators corresponding to the m level of the j th column factor. The average value of K_{jm} , denoted as k_{jm} , allows the determination of the optimal level for the j th factor and the optimal combination of all factors R_j . On the other hand, R_j stands for the range of the j th column factor, indicating the difference between the maximum and minimum values of the indicators at various levels of the j th factor. This reflects the extent of variation in the experimental indicators when the levels of the j th factor change.

A larger R_j suggests a greater impact of the factor on the experimental indicators, signifying its greater importance. Therefore, by assessing the magnitude of the range R_j , one can determine the significance of each experimental factor.

3.1. Influence of each parameter on the maximum server surface temperature

Extracting the evaluation indicator values for the maximum temperature on the server surface from Table 5 and calculating the corresponding values, from the values in the last line of Table 6, it can be seen that the magnitude of the extreme differences R for factors A, B, C and D is $R_B > R_D > R_C > R_A$. So the primary and secondary factors that have an effect on the server surface temperature are ranked BDCA. Therefore, the orthogonal experimental factors' impact sequence on the maximum temperature on the server surface is as follows: Jet pipe diameter > Jet orifice diameter > Number of holes > Jet distance.

The orthogonal experimental factors' impact on the evaluation indicators is reflected in the numerical values of the evaluation indicators corresponding to the levels of each factor. The trend chart of the influence of each level factor (different values for different parameters) on the evaluation indicators can visually demonstrate the optimal combination of parameter values of the level factors. As

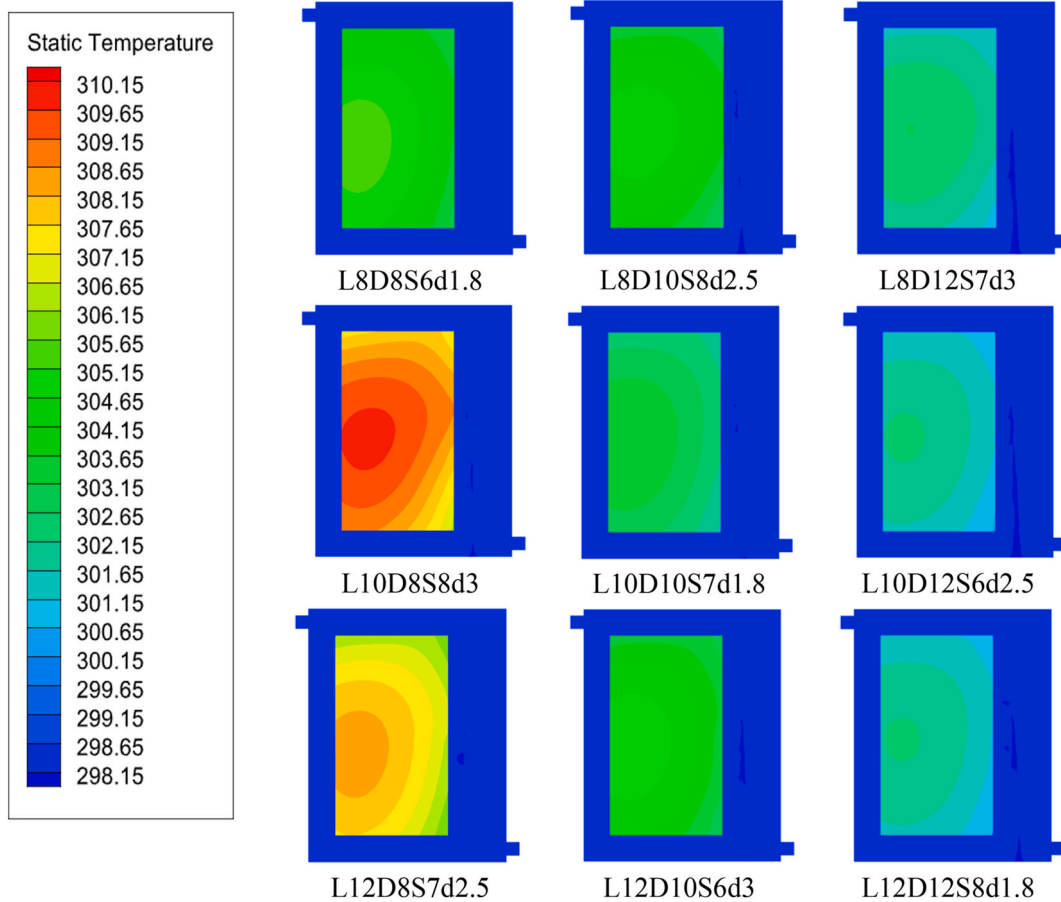


Fig. 7. The temperature distribution fields under nine simulation groups.

Table 5
Simulation results under nine simulation groups.

Group No.	Simulation indicators			
	Server Maximum Temperature T_{max} (°C)	Differential pressure between inlet and outlet ΔP (Pa)	Heat transfer coefficient h (W/m ² ·K)	Maximum pressure at characteristic point P_{max} (Pa)
1	32.2	10601	15469	8201
2	32	9955	16290.2	12108
3	30	12883	24955.1	17135.2
4	37	6011.1	9207.3	6813
5	30.4	12227.3	208263	10856
6	29.1	13340	28040	16625.5
7	35.2	6369.1	10622	6172.1
8	31.8	8308.1	16318	9500
9	29	13811.1	28074	17388

shown in Fig. 8, according to the orthogonal experiment, the optimal parameter combination is $A_1B_3C_1D_1$ if the server surface temperature is to be minimized. That is, to minimise the surface temperature of the server, the optimal parameters of the system are: jet distance of 8 mm, jet pipe diameter of 12 mm, number of holes of 6, jet orifice diameter of 1.8 mm.

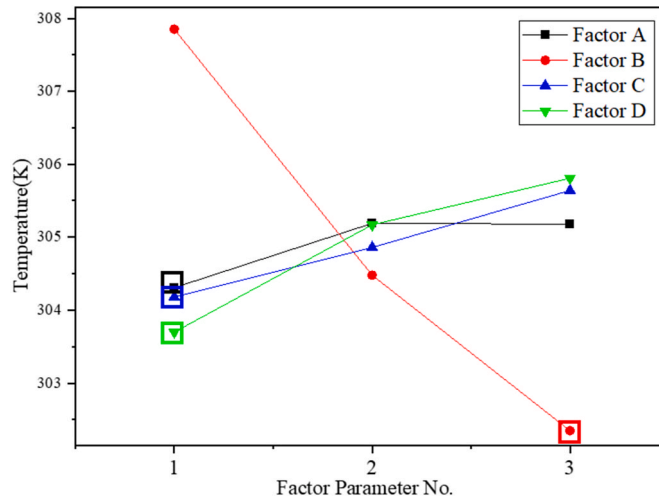
3.2. The pressure difference between the inlet and outlet of the liquid-cooled cabinet

Extracting the evaluation indicator values of the jet inlet and outlet pressure difference of the liquid-cooled cabinet from Table 5 and calculating the corresponding values, from the values in the last line of Table 7, it can be seen that the magnitude of the extreme differences R for factors A , B , C and D is $R_B > R_D > R_A > R_C$. So the primary and secondary factors affecting the pressure difference between the inlet and outlet of the liquid-cooled cabinet jet are ranked BDAC. Thus, the impact sequence of orthogonal experimental

Table 6

Range analysis table of each factor on the maximum temperature of the server surface.

Statistic	Factor A	Factor B	Factor C	Factor D
K1	93.49	104.10	93.08	91.66
K2	96.11	93.99	95.14	96.06
K3	96.09	87.59	97.46	97.97
k1	31.16	34.70	31.03	30.55
k2	32.04	31.33	31.71	32.02
k3	32.03	29.20	32.49	32.66
R	0.87	5.50	1.46	2.10

**Fig. 8.** The effect of factors on the maximum temperature of the server surface.**Table 7**

Range analysis table of each factor on the jet inlet and outlet pressure difference of the Liquid-cooled Cabinet.

Statistic	Factor A	Factor B	Factor C	Factor D
K1	33439.11	22981.76	32249.54	36640.02
K2	31578.29	30489.97	31479.42	29663.58
K3	28488.32	40033.99	29776.76	27202.10
k1	11146.37	7660.59	10749.85	12213.34
k2	10526.10	10163.32	10493.14	9887.86
k3	9496.11	13344.66	9925.59	9067.37
R	1650.26	5684.08	824.26	3145.97

factors on the jet inlet and outlet pressure difference of the liquid-cooled cabinet is as follows: Jet pipe diameter > Jet orifice diameter > Jet distance > Number of holes.

Fig. 9 shows that the best parameter combination is $A_3B_1C_3D_3$ for the liquid-cooled cabinet jet based on the orthogonal test with the minimum pressure difference between the inlet and outlet of the jet. That is, to minimise the pressure, the optimal parameters of the system are: jet distance of 12 mm, jet pipe diameter of 8 mm, number of holes of 8, jet orifice diameter of 3 mm.

3.3. Influence of parameters on heat transfer coefficient

The values of the evaluation indexes of heat transfer coefficient were extracted from Table 5, and the corresponding values were calculated. From the values in the last line of Table 8, it can be seen that the magnitude of the extreme differences R for factors A, B, C and D is $R_B > R_D > R_C > R_A$. So the primary and secondary factors affecting the heat transfer coefficient are ranked BDCA. Therefore, the impact sequence of orthogonal experimental factors on the heat transfer coefficient is as follows: Jet pipe diameter > Jet orifice diameter > Number of holes > Jet distance.

Fig. 10 shows that the best parameter combination is $A_2B_3C_1D_1$ based on the largest heat transfer coefficient of the orthogonal test. That is, to maximise the heat transfer coefficients, the optimal parameters of the system are: jet distance of 10 mm, jet pipe diameter of 12 mm, number of holes of 6, jet orifice diameter of 1.8 mm.

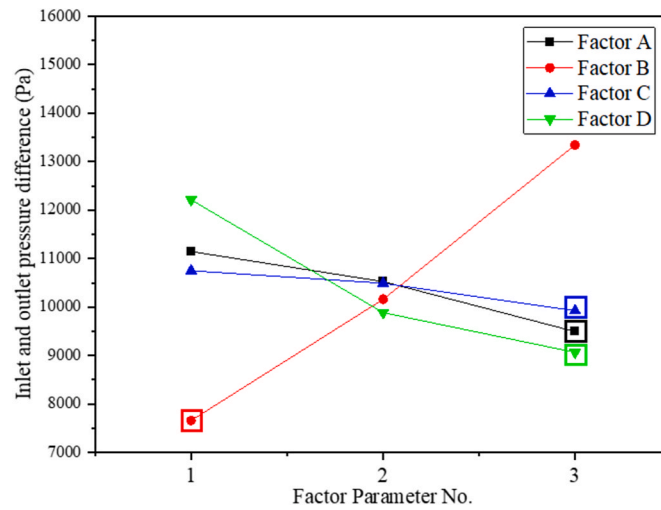


Fig. 9. The influence trend of each factor on the pressure difference between the inlet and outlet of the liquid cooling cabinet jet.

Table 8

Range analysis table of each factor on the heat transfer coefficients.

Statistic	Factor A	Factor B	Factor C	Factor D
K1	56715.23	35299.20	59827.35	64369.59
K2	58072.78	53433.49	56402.72	54951.99
K3	55013.70	81069.02	53571.65	50480.13
k1	18905.08	11766.40	19942.45	21456.53
k2	19357.59	17811.16	18800.91	18317.33
k3	18337.90	27023.01	17857.22	16826.71
R	1019.70	15256.61	2085.23	4629.82

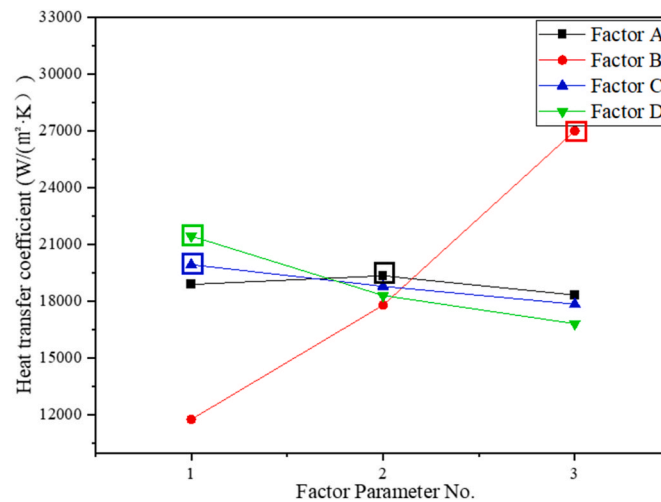


Fig. 10. The influence trend of each factor on the heat transfer coefficient.

3.4. Influence of factors on the maximum pressure at the characteristic point

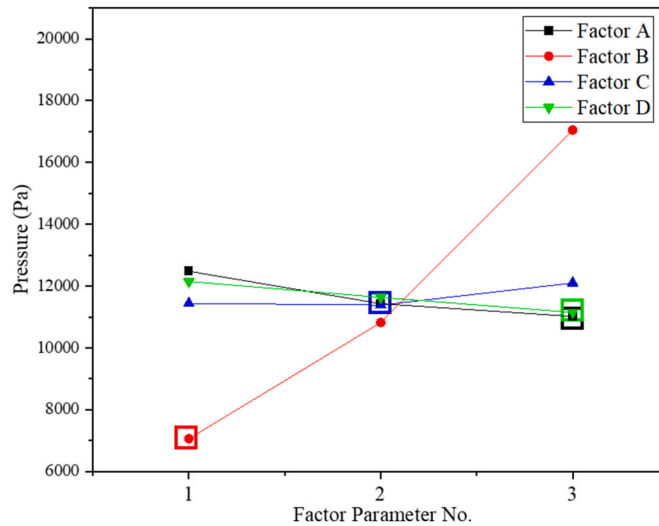
The evaluation index values of the highest pressure at the characteristic point were extracted from Table 5, and the corresponding values were calculated. From the values in the last line of Table 9, it can be seen that the magnitude of the extreme differences R for factors A, B, C and D is $R_B > R_A > R_D > R_C$. So the primary and secondary factors affecting the maximum pressure at the characteristic point are ranked BADC, meaning that the orthogonal experimental factors affecting the highest pressure characteristic points are in the following order: Jet pipe diameter > Jet distance > Jet orifice diameter > Number of holes.

Fig. 11 illustrates that, based on orthogonal experiments, the minimum highest pressure characteristic point corresponds to the

Table 9

Range analysis table of each factor on the maximum pressure at the characteristic point.

Statistic	Factor A	Factor B	Factor C	Factor D
K1	37444.24	21185.94	34326.24	36444.73
K2	34294.29	32463.74	34163.39	34905.55
K3	33059.45	51148.29	36308.35	33447.69
k1	12481.41	7061.98	11442.08	12148.24
k2	11431.43	10821.25	11387.80	11635.18
k3	11019.82	17049.43	12102.78	11149.23
R	1461.60	9987.45	714.99	999.01

**Fig. 11.** The influence trend of each factor on the maximum pressure at the characteristic point.

optimal parameter combination of $A_3B_1C_2D_3$. That is, to minimise the pressure at the characteristic point, the optimal parameters of the system are: jet distance of 12 mm, jet pipe diameter of 8 mm, number of holes of 7, jet orifice diameter of 3 mm.

Based on the impact of various parameters on the four evaluation indicators above, the structural parameters are selected as follows: jet distance $L = 8$ mm, jet pipe diameter $D = 12$ mm, number of holes $S = 6$, and jet orifice diameter $d = 1.8$ mm. This configuration results in the lowest temperature on the server surface, with the jet pipe diameter value having the most significant influence on its temperature.

The jet pipe diameter has the greatest influence on the inlet and outlet pressure difference of the liquid-cooled cabinet. To minimise the inlet and outlet pressure difference, the structural parameters should be selected as follows: jet distance $L = 12$ mm, jet pipe diameter $D = 8$ mm, number of holes $S = 8$, and jet orifice diameter $d = 3$ mm.

For the heat transfer coefficient, the jet pipe diameter will ultimately affect its value. The structural parameters are chosen as follows: jet distance $L = 10$ mm, jet pipe diameter $D = 12$ mm, number of holes $S = 6$, and jet orifice diameter $d = 1.8$ mm, resulting in the smallest temperature difference at the characteristic point.

When the structural parameters are set as follows: jet distance $L = 12$ mm, jet pipe diameter $D = 8$ mm, number of holes $S = 7$, and

Table 10

Summary of advantages and disadvantages of each parameter combination in the analyzed results.

Structural parameters	Advantages	Disadvantages
$L = 8$ mm, $D = 12$ mm, $S = 6$, $d = 1.8$ mm	Large jet flow rates and small jet orifice diameter result in better cooling effects with the lowest surface temperature on the server.	Due to the minimal jet distance, the pressure exerted on the server surface is higher, which is unfavorable for the stable operation of the server.
$L = 12$ mm, $D = 8$ mm, $S = 8$, $d = 3$ mm	Smallest pressure difference at the inlet and outlet of the jet, which leads to minimal pressure fluctuations within the liquid cooling cabinet, ensuring stable operation.	The jet pipe is too far away from the server, with a small diameter, resulting in a low flow rate of the fluorinated liquid, which leads to poor cooling effectiveness for the server.
$L = 10$ mm, $D = 12$ mm, $S = 6$, $d = 1.8$ mm	The jet pipe has a high flow rate, with small jet orifice diameter, resulting in good cooling effectiveness and a high heat transfer coefficient.	A large flow rate will lead to an increase in the cooling liquid pressure on the surface of the server, resulting in reduced system stability.
$L = 12$ mm, $D = 8$ mm, $S = 7$, $d = 3$ mm	A large distance of the jet, coupled with a large jet orifice diameter, results in a lower jet pressure on the surface of the server, which is beneficial for the stable operation of the server.	As the jet distance and jet orifice diameter increase, the cooling effect decreases accordingly.

jet orifice diameter $d = 3$ mm, the value of the highest pressure at the characteristic point is minimized. The jet pipe diameter has the greatest influence on the highest pressure at the characteristic point.

The advantages and disadvantages of the four structural parameters are listed in Table 10.

Among the optimal parameters analyzed above, the jet distance has the second-largest effect on the highest pressure at the characteristic point. To minimise the pressure at the characteristic point, it is proposed to set the jet distance at $L = 12$ mm. Simultaneously, the jet pipe diameter has the greatest impact on all parameters. A larger jet pipe diameter results in higher heat transfer coefficients and lower maximum temperature on the server surface. However, it also leads to increased inlet-outlet pressure differentials and maximum pressure at characteristic points. Considering both pressure and cooling effects, a compromise is made, and a jet pipe diameter of $D = 10$ mm is selected. When the number of holes is $S = 6$, the server's surface experiences the lowest maximum temperature and the highest heat transfer coefficient.

Additionally, the number of holes has the least impact on the inlet-outlet pressure differentials and the maximum pressure at characteristic points in the liquid cooling cabinet. Therefore, the number of nozzles are selected as $S = 6$. The jet orifice diameter has the second-largest impact on the server's surface maximum temperature, inlet-outlet pressure differentials and heat transfer coefficient. Its effect on the maximum pressure at characteristic points is the second smallest. Neglecting the influence of inlet-outlet pressure differentials, the jet orifice diameter of $d = 1.8$ mm is selected. Based on the above analysis, the optimal structural parameters are approximately determined as follows: jet distance $L = 12$ mm, jet pipe diameter $D = 10$ mm, number of holes $S = 6$ and jet orifice diameter $d = 1.8$ mm.

4. Effect of different modes of heat transfer coefficient and pressure difference

4.1. Influence of different inlet and outlet arrangements on the heat transfer coefficient

Based on the optimal structural parameters determined from the orthogonal experiment, further investigation is conducted on the effects of four inlet-outlet configurations (Z-type bottom inlet and top outlet, Z-type top inlet and bottom outlet, U-type bottom inlet and top outlet, and U-type top inlet and bottom outlet) of the liquid cooling cabinet on the flow and heat transfer characteristics, as shown in Fig. 12.

Fig. 13 and Fig. 14 illustrate the temporal evolution of the average server temperature and heat transfer coefficient under various inlet-outlet arrangements. It can be observed that as time progresses, the temperature of the server has been showing a continuous

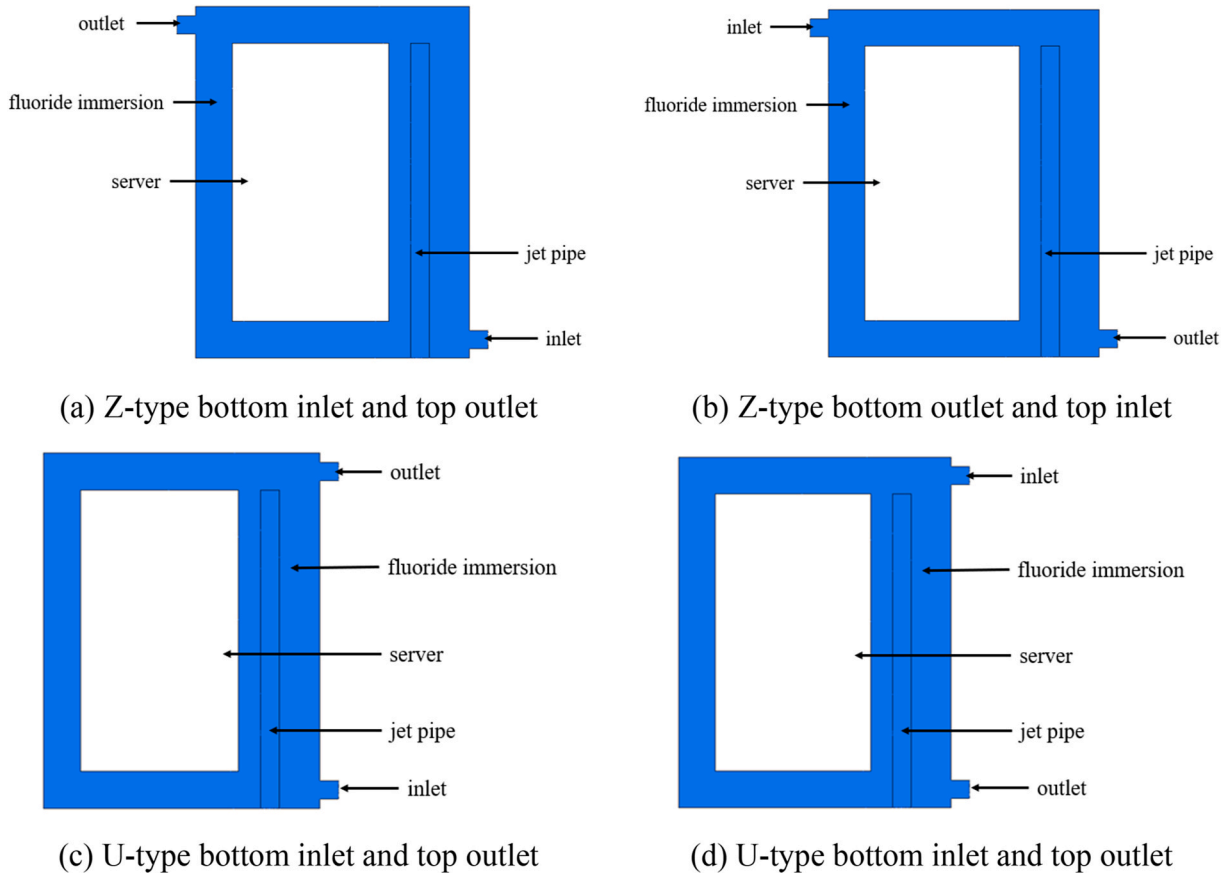


Fig. 12. Z-type and U-type inlet and outlet arrangement.

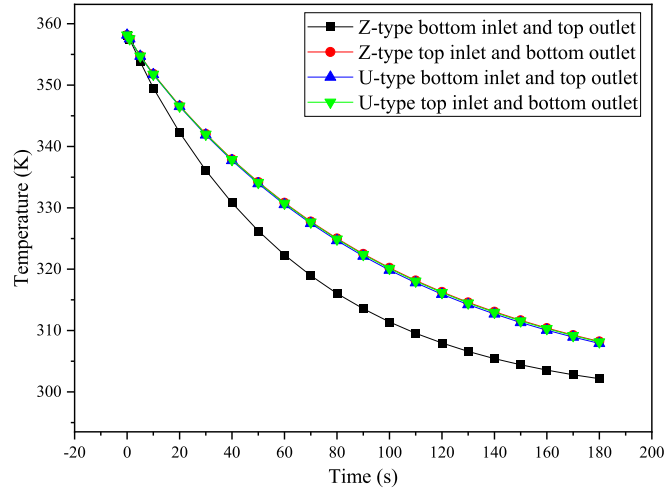


Fig. 13. Server temperature variation over time under different inlet and outlet configurations.

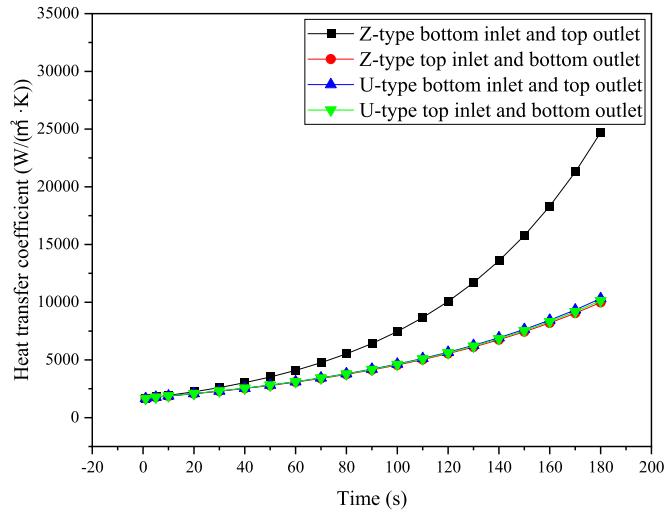


Fig. 14. Server's heat transfer coefficient over time under different inlet and outlet configurations.

downward trend, albeit with a gradually decreasing rate of decline, while the heat transfer coefficient gradually increases, with the rate of increase gradually steepening. When the liquid cooling cabinet adopts the configuration of Z-type bottom inlet and top outlet, the cooling effect is optimal. The final temperature reached by the server is as low as 29 °C, and the achievable heat transfer coefficient is the highest, reaching up to 24698 W/(m²·K).

The cooling effects under the three inlet-outlet configurations of Z-type top inlet and bottom outlet, U-type bottom inlet and top outlet, and U-type top inlet and bottom outlet are similar, with temperatures and heat transfer coefficients basically maintaining the same trend. The final temperature is 35 °C, and the heat transfer coefficients are approximately 9977.3 W/(m²·K), 10314.3 W/(m²·K), and 10132.1 W/(m²·K) respectively. When the liquid cooling cabinet adopts the configuration of Z-type bottom inlet top outlet, the heat transfer coefficient is approximately 2.5 times higher compared to the other three configurations.

4.2. Influence of different inlet and outlet arrangements on the pressure

The pressure on the surface of the server is a key physical parameter in the immersion cooling system. Attention to the pressure on the surface of the server is an important factor in guaranteeing the efficient and reliable operation of the immersion jet system. Fig. 15 illustrates the impact patterns of different inlet and outlet methods on the surface pressure of servers. Overall, the pressure at server characteristic points exhibits a fluctuating pattern. Under the same vertical arrangement of inlet and outlet, the pressure variation trends on the server surface are similar, regardless of whether a “Z” or “U” layout is used. In the design where the outlet is located above, and the inlet is below, the pressure on the server surface can be reduced by 17.1 %–31.6 %, compared to the design where the inlet is above and the outlet is below.

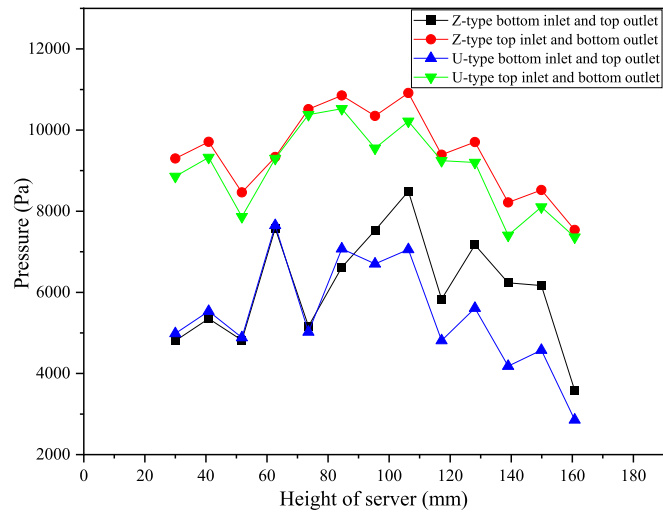
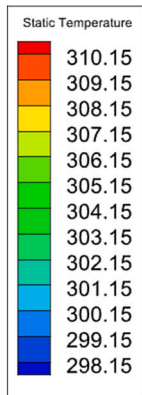


Fig. 15. Pressure at the characteristic points of the server under different inlet and outlet configurations.



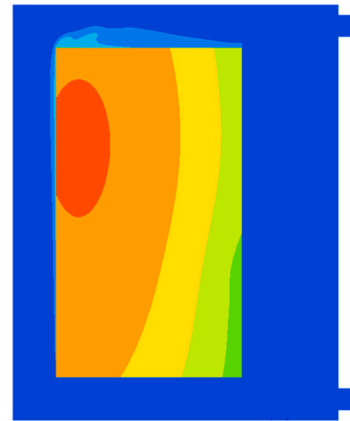
(a) Z-type bottom inlet and top outlet



(b) Z-type top inlet and bottom outlet



(c) U-type bottom inlet and top outlet



(d) U-type top inlet and bottom outlet

Fig. 16. Temperature distribution fields of four inlet and outlet arrangements.

4.3. Influence of different inlet and outlet arrangements on the uniformity of server surface temperature

Fig. 16 displays temperature distribution under different inlet and outlet configurations. The results indicate that, in the layout with bottom inlet and top outlet under the “Z” configuration, the server temperature reaches the lowest point. Following this, the “U” configuration with top inlet and bottom outlet exhibits a slightly higher temperature. Both configurations with top inlet and bottom outlet demonstrate comparable cooling effectiveness in the final server cooling outcome. In addition, under the Z-type top-inlet/bottom-outlet, U- type bottom-inlet/top-outlet, and U- type top-inlet/bottom-outlet approaches, the server surface exhibits multiple temperature stratifications. In contrast, there is less temperature stratification under the Z-type lower inlet and outlet layout, suggesting to some extent that the temperature distribution on the server surface is more uniform in this configuration. Regardless of the specific inlet and outlet layout, the server surface temperature on the right, closer to the jet pipe, is consistently lower than that on the left, farther away from the jet pipe.

Fig. 17 illustrates the velocity vectors for four inlet and outlet arrangements in which the coolant is jetted from the inlet of the jet pipe to the surface of the server at a certain flow rate and forms a vortex as it reaches the surface. In the Z-type bottom inlet and top outlet arrangement shown in Fig. 17 (a), the coolant passes through the upper and lower parts of the server before flowing towards the upper left outlet. In the U-type bottom inlet and top outlet configuration, the coolant flows directly to the right upper outlet after being ejected from the jet tube. Apart from the jetting surface of the server, other surfaces experience minimal jet disturbances. When the inlet/outlet configuration is top inlet and bottom outlet, the coolant jets onto the server surface and flows upward. Upon reaching the top of the cabinet, it then flows out towards the bottom right outlet. This also explains why, in Fig. 13, under the Z-type bottom inlet and top outlet configuration, the temperature of the server surface is the lowest. This is because in this configuration, the coolant surrounding the immersed server experiences significant jet disturbances, whereas in other configurations, apart from the server's jetting surface, coolant flow on other surfaces is less pronounced, resulting in relatively higher temperatures for the server.

According to the temperature distribution cloud diagram and velocity vector diagram, it is apparent that different inlet/outlet configurations result in varying levels of disturbance in different regions, leading to different cooling effects on the server. The server surfaces closer to the jet tube experience higher flow velocities, resulting in better cooling performance. Conversely, surfaces farther away from the jet tube exhibit more immersion cooling effects compared to jet cooling effects, leading to higher temperatures. Therefore, representative characteristic points are selected as shown in the following Figure to study the overall temperature

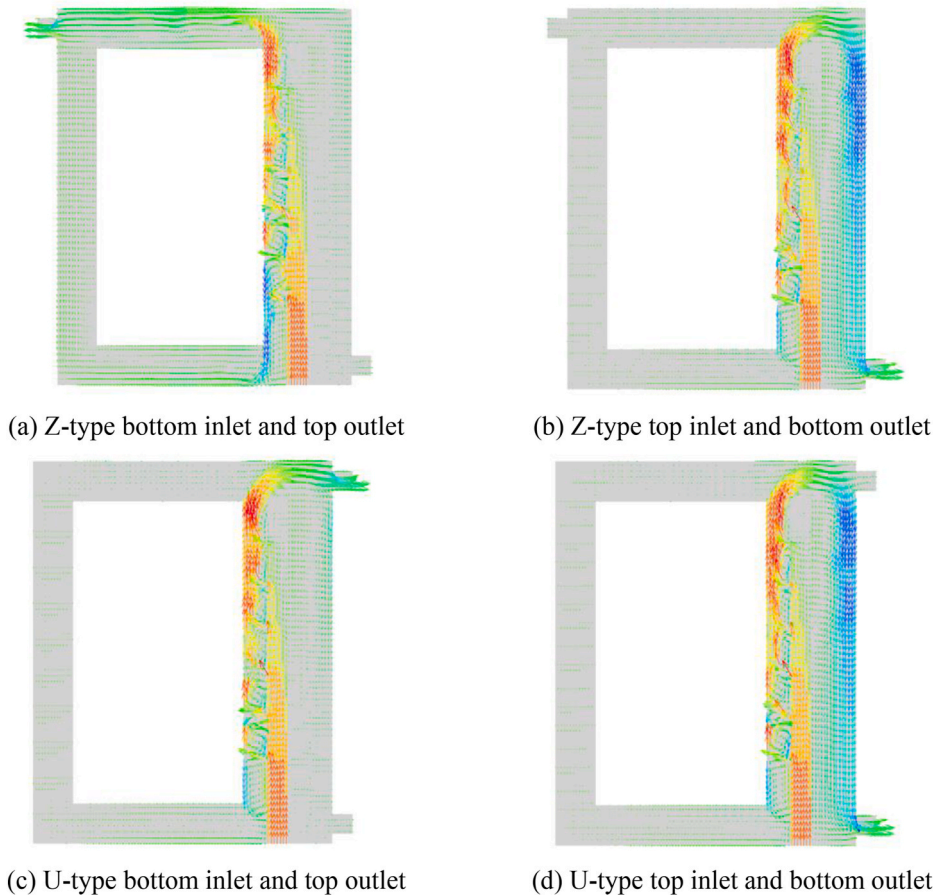


Fig. 17. Velocity vector of four inlet and outlet arrangements.

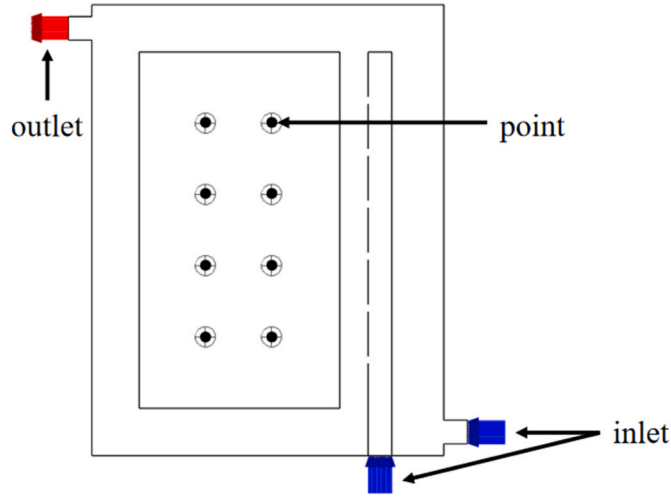


Fig. 18. Schematic of feature points.

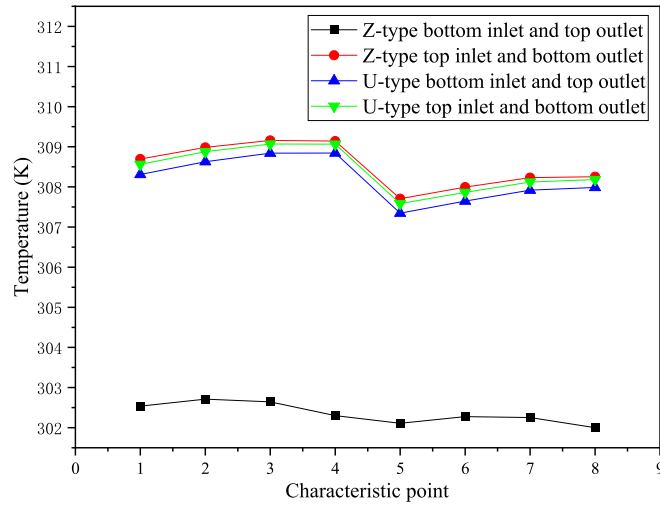


Fig. 19. Temperature changes at characteristic points under different inlet/outlet configurations.

uniformity of servers under different inlet configurations.

Fig. 19 presents the temperature variation trend at the selected feature points in Fig. 18. It can be observed from the graph that, regardless of the chosen inlet/outlet arrangement, the temperature on the left side of the server is generally higher than the right side. Under the Z-type top inlet and bottom outlet, U-type bottom inlet and top outlet, and U-type top inlet and bottom outlet configurations, the temperature variation trend at the feature points remains consistent.

It is worth noting that under the Z-type bottom inlet and top outlet layout, the temperature difference at the feature points is relatively small. In this layout, the temperature on the server surface reaches the lowest level, indicating a significant improvement in cooling efficiency with this inlet/outlet configuration. In contrast, the cooling effect of the U-type bottom inlet and top outlet is slightly lower, followed by the U-type top inlet and bottom outlet, while the Z-type top inlet and bottom outlet layout shows the worst cooling effect.

The temperature distribution fields in Fig. 16 shows that the overall temperature of the server is not uniform under different inlet/outlet arrangements, and there are temperature stratification phenomena from left to right. In order to evaluate this temperature inhomogeneity, 8 feature points were selected on the server and the temperatures of the 8 feature points were counted. The temperature standard deviation, S , was defined using the following formula to assess the overall temperature uniformity of the server.

$$S = \sqrt{\frac{1}{N} \sum_{i=1}^N (T_i - T_{ave})^2} \quad (5)$$

In the Equation:

S —Standard deviation, the defined indicator of temperature uniformity;

N —Number of characteristic point temperatures counted;

T_i —Temperature of a single characteristic point;

T_{ave} —Average of the temperatures of the characteristic points counted.

Fig. 20 illustrates the comparison of temperature uniformity between Z-type and U- type configurations in server layout. As shown in Fig. 20, under the Z-type configuration with bottom inlet and top outlet, the server exhibits the best temperature uniformity. Following this, the Z-type configuration with top inlet and bottom outlet ranks next, with the U- type configuration featuring bottom inlet and top outlet in the middle, and the U- type configuration with top inlet and bottom outlet displaying the poorest temperature uniformity.

Based on the cooling performance and temperature uniformity results, it can be concluded that by utilizing the Z-type configuration with top inlet and bottom outlet for the liquid-cooled cabinet under optimal jet impingement tube structure parameters, the immerse jet cooling performance for the servers and the temperature uniformity of the server can be further improved.

5. Dimensional analysis and dimensionless criterion relation

5.1. Dimensional analysis

The dimensional analysis method is based on the principle of dimensional harmony of equations. A mathematical equation that can perfectly and correctly reflect the physical process must be homogeneous. There are two dimensional analysis methods: one is applicable to the relationship between the influencing factors in the form of a single index, called the Rayleigh method; another universal method is called the π theorem. The π -theorem is formulated as follows: There is a physical equation $F(x_1, x_2, x_3, \dots, x_m) = 0$ with m variables. Among them, r variables can be selected as independent of each other in terms of magnitude, and the remaining $(m-r)$ variables are non-independent. This physical equation, then, can necessarily be expressed as a physical equation of $(m-r)$

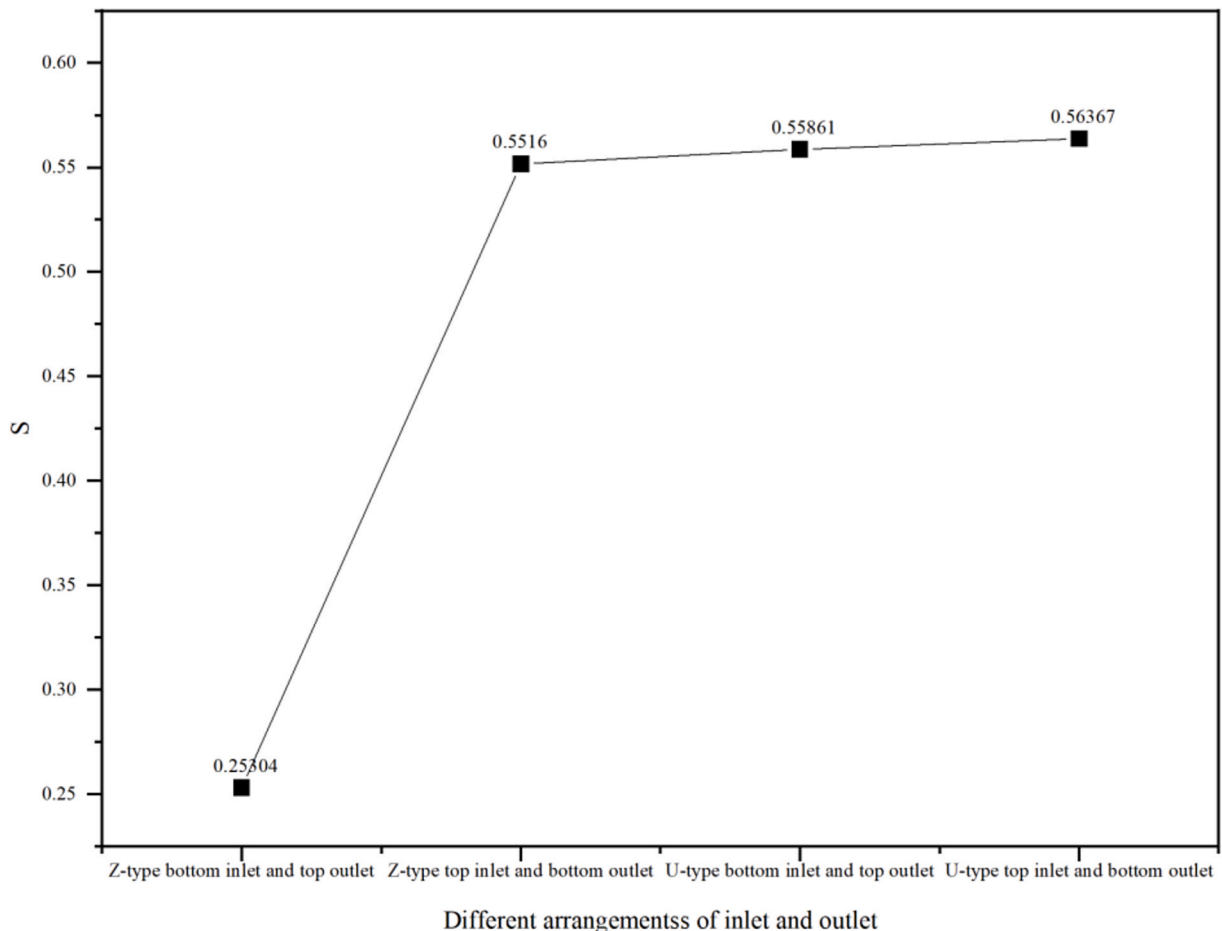


Fig. 20. Variation patterns of temperature uniformity coefficients under different inlet/outlet configurations.

dimensionless quantities, i.e., $F(\pi_1, \pi_2, \dots, \pi_{m-r}) = 0$, where $\pi_1, \pi_2, \dots, \pi_{m-r}$ are $(m-r)$ dimensionless quantities consisting of a number of independent variables consisting of dimensionless quantities.

5.1.1. Heat transfer coefficient h

The basic physical quantities in the field of heat transfer flow are: mass M , length L , time T , temperature θ . The factors affecting the heat transfer coefficient h of the submerged jet include 8 items: heat transfer coefficient h , jet distance L , jet pipe diameter D , opening area A , coolant density ρ_a , coolant temperature T_a , coolant specific heat c_p , and the local acceleration of gravity g . The names of these physical quantities, their symbols, and their magnitudes are shown in Table 11.

The expression for the heat transfer coefficient, h , affecting the submerged jet system, is calculated according to Eq. (6):

$$f(h, L, D, A, \rho_a, T_a, g, c_p) = 0 \quad (6)$$

select M, L, T, θ as the basic dimension, T_a, ρ_a, g, c_p as the basic physical quantity, then the dimensionless form of the remaining four physical quantities can be calculated according to Eq. (7):

$$\begin{cases} \pi_1 = T_a^{\alpha_1} \rho_a^{\beta_1} g^{\gamma_1} c_p^{\varepsilon_1} h = \theta^{\alpha_1} [ML^{-3}]^{\beta_1} [LT^{-2}]^{\gamma_1} [L^2 T^{-2} \theta^{-1}]^{\varepsilon_1} MT^{-3} \theta^{-1} \\ \pi_2 = T_a^{\alpha_2} \rho_a^{\beta_2} g^{\gamma_2} c_p^{\varepsilon_2} L = \theta^{\alpha_2} [ML^{-3}]^{\beta_2} [LT^{-2}]^{\gamma_2} [L^2 T^{-2} \theta^{-1}]^{\varepsilon_2} L \\ \pi_3 = T_a^{\alpha_3} \rho_a^{\beta_3} g^{\gamma_3} c_p^{\varepsilon_3} D = \theta^{\alpha_3} [ML^{-3}]^{\beta_3} [LT^{-2}]^{\gamma_3} [L^2 T^{-2} \theta^{-1}]^{\varepsilon_3} L \\ \pi_4 = T_a^{\alpha_4} \rho_a^{\beta_4} g^{\gamma_4} c_p^{\varepsilon_4} A = \theta^{\alpha_4} [ML^{-3}]^{\beta_4} [LT^{-2}]^{\gamma_4} [L^2 T^{-2} \theta^{-1}]^{\varepsilon_4} L^2 \end{cases} \quad (7)$$

Based on the principle of dimensional harmony, the coefficients of each equation can be obtained by solving the equation, as shown in Eq. (8):

$$\begin{cases} \alpha_1 = -1/2, \beta_1 = -1, \gamma_1 = 0, \varepsilon_1 = -3/2 \\ \alpha_2 = -1, \beta_2 = 0, \gamma_2 = 1, \varepsilon_2 = -1 \\ \alpha_3 = -1, \beta_3 = 0, \gamma_3 = 1, \varepsilon_3 = -1 \\ \alpha_4 = -2, \beta_4 = 0, \gamma_4 = 2, \varepsilon_4 = -2 \end{cases} \quad (8)$$

Then, the dimensionless π parameters of the remaining four physical quantities are, as shown in Eq. (9):

$$\begin{cases} \pi_1 = \frac{h}{T_a^{1/2} \rho_a c_p^{3/2}} \\ \pi_2 = \frac{gL}{T_a c_p} \\ \pi_3 = \frac{gD}{T_a c_p} \\ \pi_4 = \frac{g^2 A}{T_a^2 c_p^2} \end{cases} \quad (9)$$

The dimensionless equation of heat transfer coefficient h can be expressed as Eq. (10):

$$\begin{aligned} f\left(\frac{h}{T_a^{1/2} \rho_a c_p^{3/2}}, \frac{gL}{T_a c_p}, \frac{gD}{T_a c_p}, \frac{g^2 A}{T_a^2 c_p^2}\right) &= 0 \\ \Rightarrow \frac{h}{T_a^{1/2} \rho_a c_p^{3/2}} &= f\left(\frac{DT_a^2 c_p^2}{LAg^2}\right) \end{aligned} \quad (10)$$

Define the dimensionless heat transfer coefficient: $h^* = \frac{h}{T_a^{1/2} \rho_a c_p^{3/2}}$, the ratio of dimensionless jet ratio (jet diameter ratio jet distance):

$D^* = \frac{D}{L}$, dimensionless opening area: $A^* = \frac{Ag^2}{T_a^2 c_p^2}$.

Finally, the dimensionless equation of heat transfer coefficient $h \times$ can be expressed by Eq. (11):

Table 11
Physical quantities affecting the heat transfer coefficient of submerged jet systems.

Serial No.	physical quantity	notation	dimension	Serial No.	physical quantity	notation	dimension
1	heat transfer coefficient	h	$MT^{-3}\theta^{-1}$	5	coolant density	ρ_a	ML^{-3}
2	jet distance	L	L	6	coolant temperature	T_a	θ
3	jet pipe diameter	D	L	7	acceleration of gravity	g	LT^{-2}
4	opening area	A	L^2	8	coolant specific heat	c_p	$L^2 T^{-2} \theta^{-1}$

$$h^* = f\left(\frac{D^*}{A^*}\right) \quad (11)$$

The relationship between dimensionless heat transfer coefficient h^* and dimensionless jet ratio D^* and dimensionless opening area A^* is calculated by formula (11), and fitted by a quadratic function. The results are shown in Fig. 21.

It can be seen from Fig. 21 that the correlation coefficient R^2 of the curve is 80 %, and the curve fitting degree is good. The prediction model of dimensionless heat transfer coefficient h^* can be described by Eq. (12):

$$h^* = -0.14\left(\frac{D^*}{A^*}\right)^2 + 1.41\frac{D^*}{A^*} - 1.18 \quad (12)$$

The applicable conditions are: $D^*/A^* = 1.5\text{--}7.5$.

5.1.2. Server pressure P

The basic physical quantities in the field of heat transfer flow are: mass M , length L , time T , temperature θ . The factors affecting the server pressure P of the submerged jet include 8 items: server pressure P , jet distance L , jet pipe diameter D , opening area A , coolant density ρ_a , coolant temperature T_a , coolant specific heat c_p , and the local acceleration of gravity g . The names of these physical quantities, their symbols, and their magnitudes are shown in Table 12.

The expression for the server pressure P , affecting the submerged jet system, is calculated according to Eq. (13):

$$f(P, L, D, A, \rho_a, T_a, g, c_p) = 0 \quad (13)$$

select M, L, T, θ as the basic dimension, T_a, ρ_a, g, c_p as the basic physical quantity, then the dimensionless form of the remaining four physical quantities can be calculated according to Eq. (14):

$$\begin{cases} \pi_1 = T_a^{\alpha_1} \rho_a^{\beta_1} g^{\gamma_1} c_p^{\epsilon_1} P = \theta^{\alpha_1} [ML^{-3}]^{\beta_1} [LT^{-2}]^{\gamma_1} [L^2 T^{-2} \theta^{-1}]^{\epsilon_1} ML^{-1} T^{-2} \\ \pi_2 = T_a^{\alpha_2} \rho_a^{\beta_2} g^{\gamma_2} c_p^{\epsilon_2} L = \theta^{\alpha_2} [ML^{-3}]^{\beta_2} [LT^{-2}]^{\gamma_2} [L^2 T^{-2} \theta^{-1}]^{\epsilon_2} L \\ \pi_3 = T_a^{\alpha_3} \rho_a^{\beta_3} g^{\gamma_3} c_p^{\epsilon_3} D = \theta^{\alpha_3} [ML^{-3}]^{\beta_3} [LT^{-2}]^{\gamma_3} [L^2 T^{-2} \theta^{-1}]^{\epsilon_3} L \\ \pi_4 = T_a^{\alpha_4} \rho_a^{\beta_4} g^{\gamma_4} c_p^{\epsilon_4} A = \theta^{\alpha_4} [ML^{-3}]^{\beta_4} [LT^{-2}]^{\gamma_4} [L^2 T^{-2} \theta^{-1}]^{\epsilon_4} L^2 \end{cases} \quad (14)$$

Based on the principle of dimensional harmony, the coefficients of each equation can be obtained by solving the equation, as shown in Eq. (15):

$$\begin{cases} \alpha_1 = -1, \beta_1 = -1, \gamma_1 = 0, \epsilon_1 = -1 \\ \alpha_2 = -1, \beta_2 = 0, \gamma_2 = 1, \epsilon_2 = -1 \\ \alpha_3 = -1, \beta_3 = 0, \gamma_3 = 1, \epsilon_3 = -1 \\ \alpha_4 = -2, \beta_4 = 0, \gamma_4 = 2, \epsilon_4 = -2 \end{cases} \quad (15)$$

Then, the dimensionless π parameters of the remaining four physical quantities are, as shown in Eq. (16):

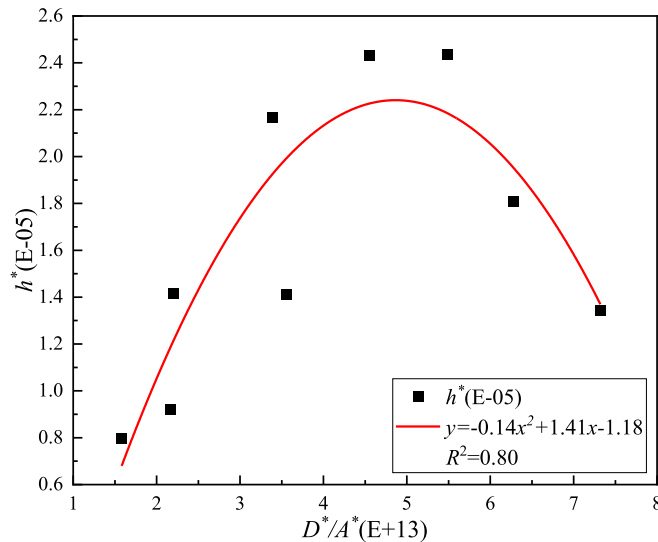


Fig. 21. Dimensionless heat transfer coefficient h^* .

Table 12

Physical quantities affecting the server pressure of submerged jet systems.

Serial No.	physical quantity	notation	dimension	Serial No.	physical quantity	notation	dimension
1	server pressure	P	$\text{ML}^{-1}\text{T}^{-2}$	5	coolant density	ρ_a	ML^{-3}
2	jet distance	L	L	6	coolant temperature	T_a	θ
3	jet pipe diameter	D	L	7	acceleration of gravity	g	LT^{-2}
4	opening area	A	L^2	8	coolant specific heat	c_p	$\text{L}^2\text{T}^{-2}\theta^{-1}$

$$\begin{cases} \pi_1 = \frac{P}{T_a \rho_a c_p} \\ \pi_2 = \frac{gL}{T_a c_p} \\ \pi_3 = \frac{gD}{T_a c_p} \\ \pi_4 = \frac{g^2 A}{T_a^2 c_p^2} \end{cases} \quad (16)$$

The dimensionless equation of heat transfer coefficient h can be expressed as Eq. (17):

$$\begin{aligned} f\left(\frac{P}{T_a \rho_a c_p}, \frac{gL}{T_a c_p}, \frac{gD}{T_a c_p}, \frac{g^2 A}{T_a^2 c_p^2}\right) &= 0 \\ \Rightarrow \frac{P}{T_a \rho_a c_p} &= f\left(\frac{DT_a^2 c_p^2}{LA g^2}\right) \end{aligned} \quad (17)$$

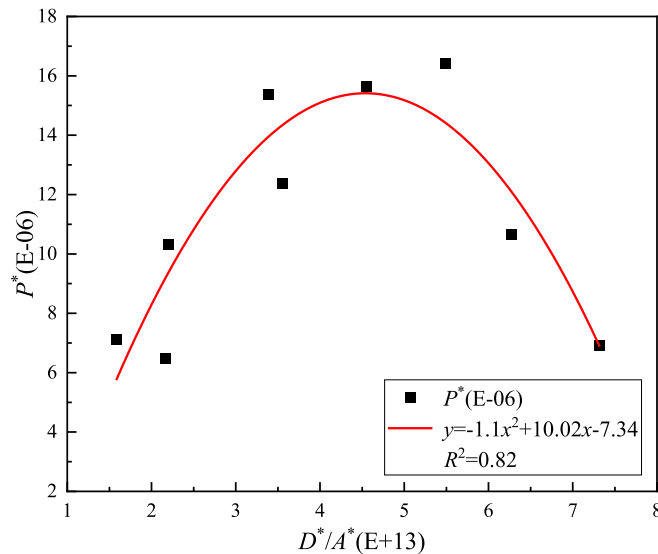
Define the dimensionless server pressure: $P^* = \frac{P}{T_a \rho_a c_p}$, the ratio of dimensionless jet ratio: $D^* = \frac{D}{L}$, dimensionless opening area: $A^* = \frac{Ag^2}{T_a^2 c_p^2}$.

Finally, the dimensionless equation of dimensionless server pressure P^* can be expressed by Eq. (18):

$$P^* = f\left(\frac{D^*}{A^*}\right) \quad (18)$$

The relationship between dimensionless server pressure P^* and dimensionless jet ratio D^* and dimensionless opening area A^* is calculated by formula (18), and fitted by a quadratic function. The results are shown in Fig. 22.

It can be seen from Fig. 21 that the correlation coefficient R^2 of the curve is 82 %, and the curve fitting degree is good. The prediction model of dimensionless server pressure P^* can be described by Eq. (19):

**Fig. 22.** Dimensionless server pressure P^* .

$$P^* = -1.11 \left(\frac{D^*}{A^*} \right)^2 + 10.02 \frac{D^*}{A^*} - 7.34 \quad (19)$$

The applicable conditions are: $D^*/A^* = 1.5\text{--}7.5$.

5.2. Dimensionless criterion relation

In the study, the dimensionless criterion number is introduced and the criterion correlation is fitted, which can more comprehensively and specifically express the flow characteristics and heat transfer characteristics in the process of submerged jet cooling server under different working conditions, which is helpful to analyze the optimization scheme of submerged jet cooling under different conditions. The dimensionless numbers related to the immersion jet cooling mainly include the Nusselt number Nu , the Reynolds number Re , the Prandtl number Pr , and the Euler number Eu . Since this study only studies the fluoride solution and does not involve other coolants, the Prandtl number Pr is a fixed value, and this study comprehensively considers the heat transfer effect and the pressure of the server. Therefore, the criterion relationship between the Nu , Re and Eu is fitted, that is, the criterion relationship is $Nu = f(Re, Eu)$.

5.2.1. Nusselt number Nu

Nu is a dimensionless parameter in fluid dynamics, which describes the relative importance of viscosity and inertial force in fluid, especially in liquid fluid. Its definition is shown in Eq. (20):

$$Nu = \frac{hl}{\lambda} \quad (20)$$

In Eq. (20), h represents the heat transfer coefficient between the fluid and the solid interface, $W/(m^2 \cdot K)$; l represents width of the server, m ; λ represents the thermal conductivity of the fluid, $W/(m \cdot K)$.

5.2.2. Reynolds number Re

Re represents the ratio of the inertial force to the viscous force of the fluid, and its definition is shown in Eq. (21):

$$Re = \frac{\rho vl}{\mu} \quad (21)$$

In Eq. (21), ρ represents the density of the fluid, kg/m^3 ; v represents the velocity of the fluid, m/s ; l represents the diameter of the jet hole, m ; μ represents the dynamic viscosity of the fluid, $Pa \cdot s$.

5.2.3. Euler number Eu

Eu represents the ratio of the pressure of the fluid to the inertial force, and its definition is shown in Eq. (22):

$$Eu = \frac{P}{\rho v^2} \quad (22)$$

In Eq. (22), P is fluid pressure, Pa ; ρ is fluid density, kg/m^3 ; v is the fluid velocity, m/s .

By analysing and fitting a limited amount of simulated data, the fitting correlation is shown in Eq. (23):

$$Nu = 57.8Re^{0.78}Eu^{-0.63} \quad (23)$$

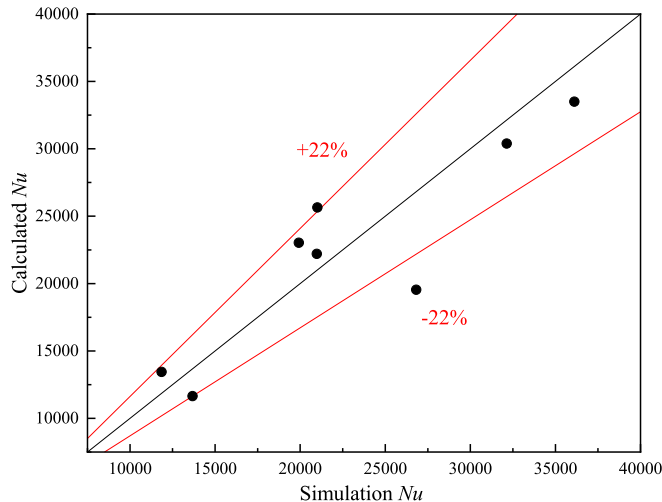


Fig. 23. The comparison of fitted values to simulated values when fitting the correlation using Re and Eu .

The applicable conditions are: $Re = 2400 \sim 5200, Eu = 1.2 \sim 3$.

Fig. 23 shows a comparison of the fitted values with the simulated values when fitting the correlation equation with Re and Eu , from which it can be seen that most of the fitted computed points fall within the $\pm 22\%$ error line of the simulated points.

6. Conclusions

Integrating the advantages of immersion liquid cooling and jet cooling holds promising prospects for future data center cooling systems. In this study, a two-dimensional model of an immersion jet-cooled server was established, and the optimal heat dissipation parameters for the immersion jet structure were obtained using orthogonal experimental methods. By varying the inlet and outlet positions of the cabinet, further analysis was conducted on the temperature and velocity fields. The research findings are as follows:

- (1) As the jet pipe diameter increases and the jet orifice diameter decreases, the server temperature decreases. As the jet distance decreases, the pressure on the server surface increases. When the diameter of the jet pipe increases by 25 %, the maximum heat transfer coefficient can be increased by approximately 60 %. When the diameter of the jet pipe increases by 50 %, the maximum heat transfer coefficient can be increased by approximately two times. Taking into account both heat transfer effectiveness and pressure considerations, the optimal structural parameters are determined through orthogonal method as follows: the ratio of jet distance (L) to jet pipe diameter (D) is 1.2, and the ratio of jet hole spacing to jet orifice diameter is 11.1;
- (2) When the outlet of the liquid-cooled cabinet is arranged at the top, the immersion jet liquid-cooled cabinet exhibits better heat dissipation performance for the servers. Specifically, when the cabinet adopts the Z-shaped configuration with bottom inlet and top outlet, and the U-shaped configuration with bottom inlet and top outlet, the final temperatures reached by the servers are 29°C and 34.8°C , respectively. Both temperatures are lower than the Z-shaped configuration with top inlet and bottom outlet at 35.1°C , and the U-shaped configuration with top inlet and bottom outlet at 35.0°C ; When the inlet and outlet of the cabinet are in a Z-shaped layout, the pressure on the server surface is reduced by 17.1 %–31.6 % compared to adopting a U-shaped layout;
- (3) When the cabinet inlet and outlet are arranged in a Z-shaped configuration with bottom inlet and top outlet, the highest heat transfer coefficient is over $240000 \text{ W}/(\text{m}^2\cdot\text{K})$. Compared to other three distributions, the heat transfer coefficient can be increased by approximately 1.5 times. Additionally, in this layout, the server surface heat dissipation is the most uniform.
- (4) According to the π theorem, the relationship between the heat transfer coefficient h , the server pressure P and the jet pipe diameter D , the jet distance L and the opening area A is obtained by dimensional analysis, respectively: $h^* = -0.14 \left(\frac{D^*}{A^*} \right)^2 + 1.41 \frac{D^*}{A^*} - 1.18$ ($D^*/A^* = 1.5 \sim 7.5$), $P^* = -1.11 \left(\frac{D^*}{A^*} \right)^2 + 10.02 \frac{D^*}{A^*} - 7.34$ ($D^*/A^* = 1.5 \sim 7.5$); according to the relevant simulation data in the third section, the fitting of the criterion correlation is carried out, and the obtained criterion correlation is: $Nu = 57.8Re^{0.78}Eu^{-0.63}$ ($Re = 2400 \sim 5200, Eu = 1.2 \sim 3$).

CRedit authorship contribution statement

Yingying Zhang: Writing – original draft, Investigation. **Yu Wang:** Writing – review & editing, Validation. **Risto Kosonen:** Writing – review & editing. **Dawar Asfandiyar Khan:** Validation. **Xincheng Wang:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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